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EDITORIAL

Performance Management in Elite Football: A Teamwork Modeling Approach

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Winning matches and getting 3 competition points is commonly established as a key performance indicator (KPI) for elite football (soccer) teams to achieve their objectives during the season. While winning is the responsibility of the football players, sport-science and -medicine practitioners' responsibility is to support them by increasing their availability for matches (a key contributing factor to winning the games and ensuring team success).[1,2] Unfortunately, it is not uncommon to observe sport-science (eg, strength and conditioning coach, physiologist) and -medicine (e.g., physician, physiotherapist) practitioners working in silos in a football setting. As such, we are proposing a teamwork modeling approach for practitioners with elite football teams. Our model comprises specific KPIs to enhance players' performance while reducing injury risk (Figure 1). This model has the potential to attract the attention of practitioners and researchers aiming to prevent injuries and improving players' recovery for enhanced football performance.

The Model

Football teams are mainly using external training-load monitoring (ie, GPS data)[3] and the acute-to-chronic–workload ratio to manage performance and injury risk.[4] However, the effectiveness and accuracy of the latter method has been questioned due to flaws in its statistical properties.[5] Moreover, understanding whether a positive or negative training effect is acute or chronic is context dependent. Indeed, a positive effect resulting from a resistance-training session (ie, increased muscle protein synthesis)[6] is accompanied by a negative effect (ie, neuromuscular fatigue).[7] Acute and chronic should be seen more as continuum/intertwined variables than a strict dichotomy.[8] Monitoring the training load in the context of athletic training is an important strategy to examine whether or not the desired training response and adaptations are elicited.[9] In our model, training-load monitoring is not adopted as an isolated approach to manage performance and injury risk; instead, it is part of a broad and holistic strategic plan composed of 4 pillars (Figure 1).

Pillar 1: Establishing a Systematic Monitoring System

Pillar 1 (P1) is central in our model due to its capacity to integrate data derived from 3 different sources: P1a, training-load monitoring; P1b, functional assessments; and P1c, injury and illness surveillance (Figure 1). The main objective of P1 is to generate sensitive information to enhance decision making process.

- *P1a—Training-Load Monitoring:* We recommend the conceptual framework of physical training proposed by Jeffries et al,[8] which allows practitioners to track the acute and chronic effects (eg, positive/negative) of the external load imposed during the training sessions on athletes' response (ie, internal load).
- *P1b—Functional Assessments:* The countermovement-jump (CMJ) test seems to be an efficient approach to quantify neuromuscular fatigue in football as an acute negative effect of training, as well as to measure the chronic effect of training on lower-body power or jumping performance.[8] Jump height, which is the most common variable examined by practitioners during a CMJ test, is, however, limited to elicit such information. Time-based metrics (eg, reactive strength index, time to takeoff, propulsive-phase duration, and time to peak power) should be used to identify neuromuscular fatigue, while output metrics (eg, jump height, peak power, mean propulsive force, propulsive impulse) are useful performance outcomes.[10]
- *P1c—Injury and Illness Surveillance:* This allows tracking injury and illness incidence and burden, informing whether the applied practices are effective in keeping players injury free and decreasing their unavailability. Daily recorded data should be structured in agreement with the IOC consensus statement.[11]

Pillar 2: Establishing the Appropriate Training Process

The main objective of Pillar 2 (P2) is to optimize training (Figure 1). Prior to training prescription, practitioners should consider (1) the performance model adopted by the team, (2) contextual and player individual factors, (3) training adaptations, (4) previous training load experienced by the players, and (5) coach experience.[8] For example, training-loading pattern should be customized for players who are more exposed to game load (ie, starters) versus those less exposed (ie, nonstarters).[12] This approach will promote optimizing performance and reducing injury risks, particularly during congested match fixtures.[12]

Pillar 3: Establishing an Evidence-Based-Practice Concept

For Pillar 3 (P3), a variety of practices including gym/field training (P3a), recovery modalities (P3b), injury prevention (P3c), and specific training following injury (P3d) are commonly applied within a football setting (Figure 1). Whenever applicable, scientific evidence should be used to decide on which practice to employ. For instance, whether adopting exercise A or B for improving lower limb power or whether rule A or B improves aerobic endurance during small-sided games. Evidence-based practice allows

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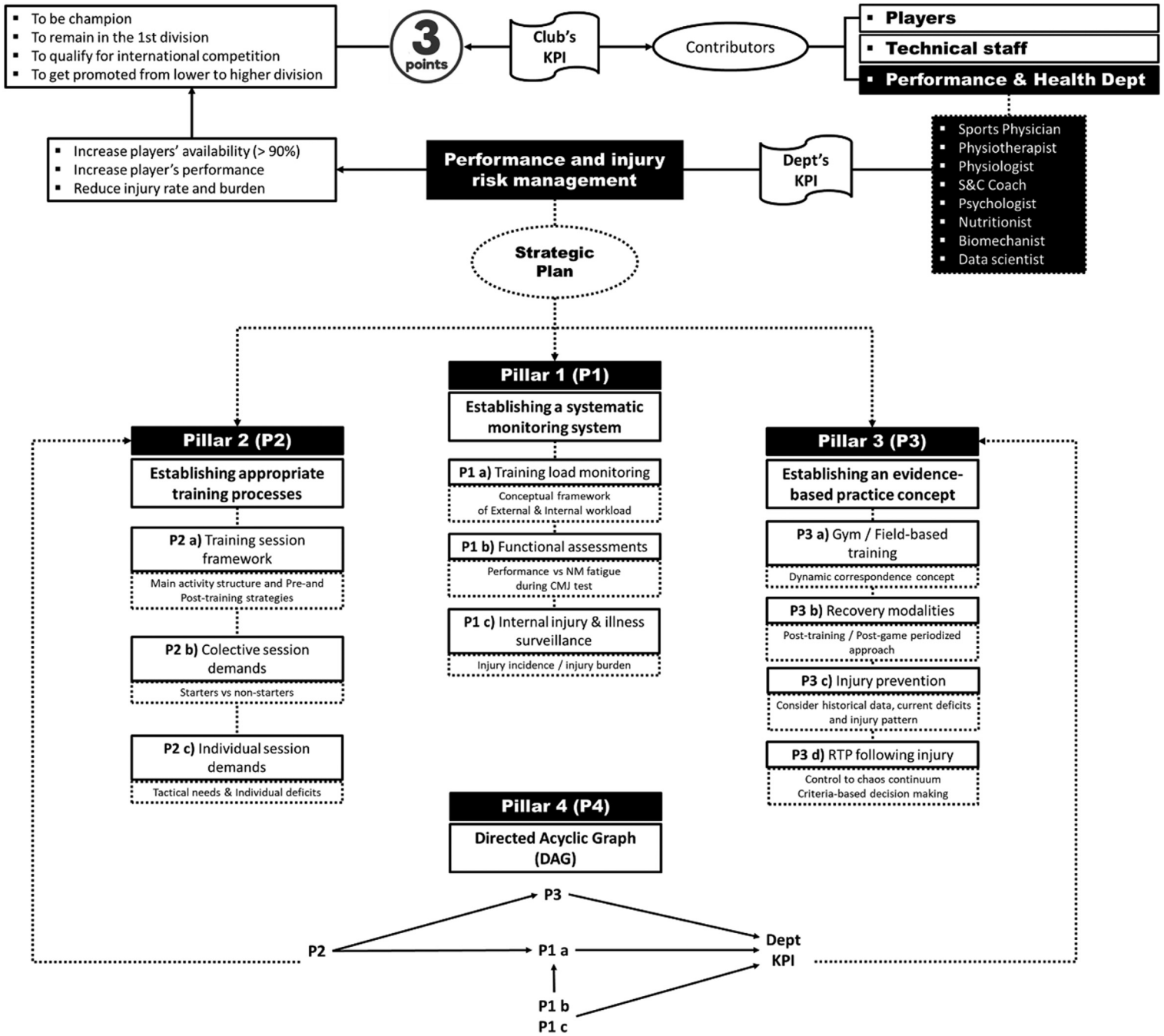

Figure 1 — Model developed to optimize players' performance while reducing injury risk for elite football teams. Abbreviations: KPI, key performance indicators; S&C, strength and conditioning; RTP, return to play; NM, neuromuscular; CMJ, countermovement jump.

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practitioners to stay abreast of the current literature and ensure alignment with recent scientific findings.[13] The number of existing studies, the consistency of their findings, and their quality should determine the strength of certainty to use for a given practice.[13]

Pillar 4: Directed Acyclic Graph

Pillar 4 (P4) adopts the directed acyclic graph (DAG), which is an effective tool to enhance teamwork and support decision-making process.[14] For instance, P1a acts as surrogate outcome of P2 and measures its effect on KPI (players' availability) (Figure 1). In addition, P3 acts as a mediator of P2 and influences their relationship. Practitioners should systematically evaluate the processes and practices adopted and modify them if necessary. Note that the DAG also promotes higher-quality internal communication between sport-science and -medicine practitioners and technical staff. These communications are a key factor to increase players' availability and reduce injury incidence in football.[15]

Final Considerations

The model proposed may have a direct practical application to enhancing football performance and should benefit sport and exercise physiologists, sport scientists, sport physicians, and sport-performance researchers currently working in an elite football environment. Future research is needed to examine the contribution of the 4 pillars and their respective components to the club's KPI (3 competition points).

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